

Multi-View Stereo Problems

- Depth map-based MVS algorithms estimate the reference view depth maps using multiple RGB inputs (Reference + Source views)
- A consistent scene requires geometric consistency of depth estimates across multiple views

Two broader approaches are undertaken to ensure geometric consistency in estimated depth maps:

- Repeated application of geometric constraints during the depth estimation process → Traditional MVS Algorithms
- Geometric constraints applied as a post-processing step → Learning-based MVS Algorithms

GC-MVSNet is a learning-based algorithm with geometric constraints applied during the learning process.

Learning-Based MVS Algorithms

A learning-based MVS method:

- Extracts multi-level features using CNNs
- Creates a matching 3D cost volume using features
- Regularizes cost volume using 3D-CNN
- Filters geometrically consistent points to generate 3D point-cloud

They only use Geometric Constraints as a post-processing step for filtering multi-view consistent points. This leads to:

- Limited geometric cues during the learning process
- Requiring more training iterations to learn to reason about geometry

Hypothesis

GC-MVSNet:

- Explicitly models cross-view geometric constraints during learning
- Penalizes geometrically inconsistent estimates during learning

With such explicit geometric constraint modeling, GC-MVSNet should:

- Develop a better understanding of multi-view geometry → Improved quantitative results
- Learn quickly to reason about scene geometry → Require less training iterations

Forward-Backward-Reprojection

Inputs: $D_0, c_0, D_i^{gt}, c_i^{gt}$
Output: D_{P_0}''', P_{P_0}'''

$K_R, E_R \leftarrow c_0; K_S, E_S \leftarrow c_i^{gt}$
 $D_{(R \rightarrow S)} \leftarrow K_S \cdot E_S \cdot E_R^{-1} \cdot K_R^{-1} \cdot D_0$ ▷ Project
 $X_{D_{(R \rightarrow S)}}, Y_{D_{(R \rightarrow S)}} \leftarrow \tilde{D}_{(R \rightarrow S)}$

$D_{S_{remap}} \leftarrow REMAP(D_i^{gt}, X_{D_{(R \rightarrow S)}}, Y_{D_{(R \rightarrow S)}})$ ▷ Remap
 $D_{P_0}'' \leftarrow K_R \cdot E_R \cdot E_S^{-1} \cdot K_S^{-1} \cdot D_{S_{remap}}$ ▷ Back project
 $P_{P_0}'' \leftarrow (X_{D_{P_0}''}, Y_{D_{P_0}''})$

Other Modifications

Two additional modifications were to stabilize the model's performance.

- Kept the feature-extraction network as a Feature Pyramid Network, replaced the regular conv-layers with deformable conv-layers
- Replaced BatchNorm-layers with GroupNorm-layers as BatchNorm is not well suited for small batch-sizes

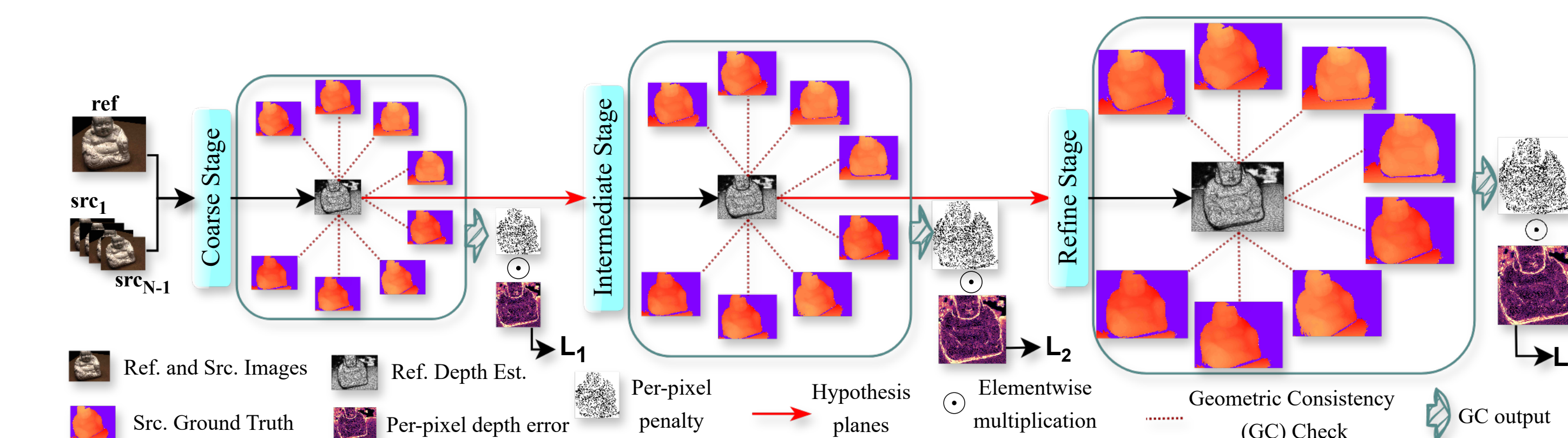
Reconstructed Scene Point Clouds



Method

Geometric-Consistency (GC) Module:

- Applied at the end of each stage to check cross-view consistency of the reference view depth maps
- Generates penalty for geometrically inconsistent estimates for each stage



Geometric-Consistency Module

Complete GC-Algorithm

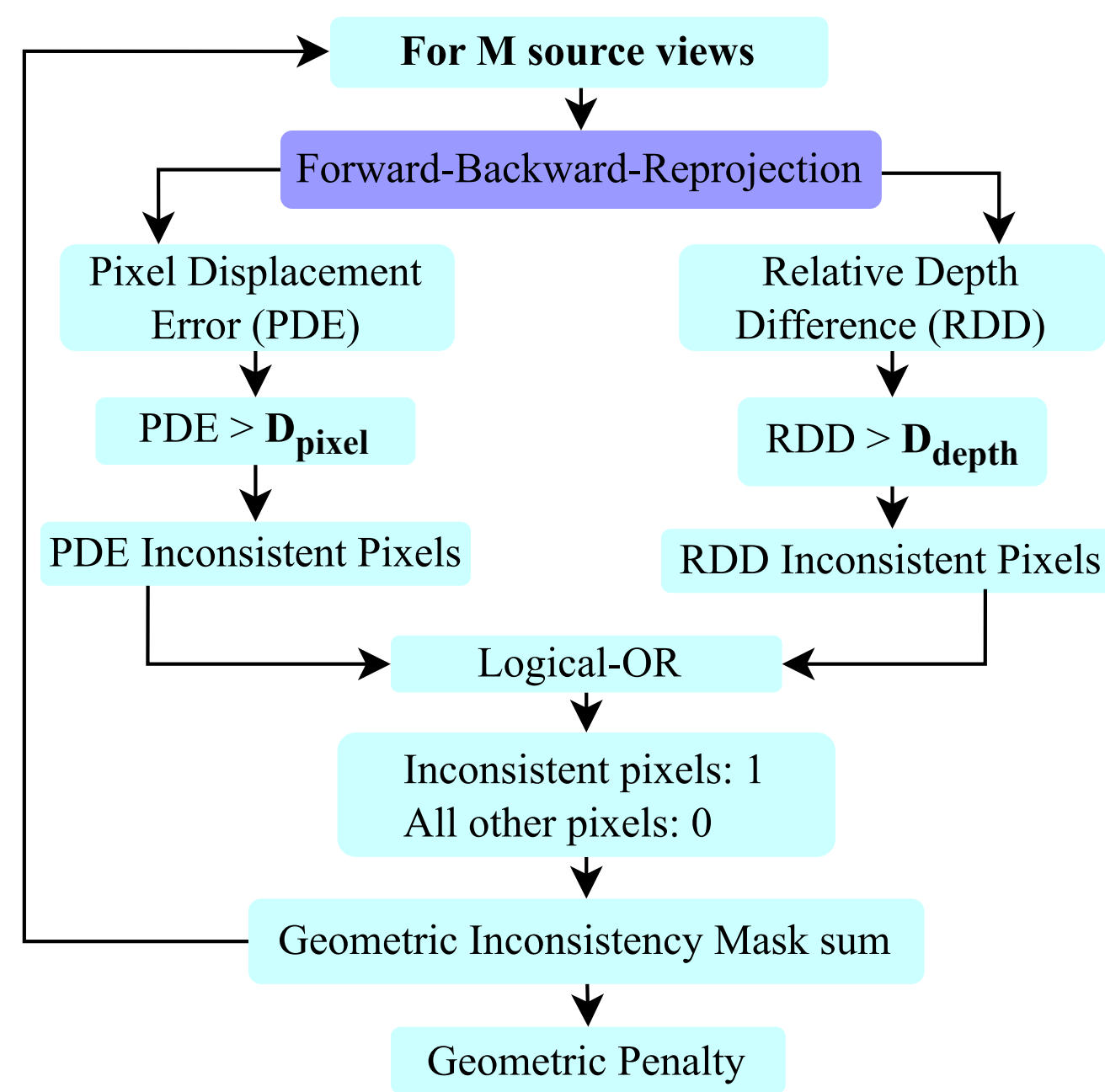
Initialize Mask-Sum → 0

For each Src. depth map:

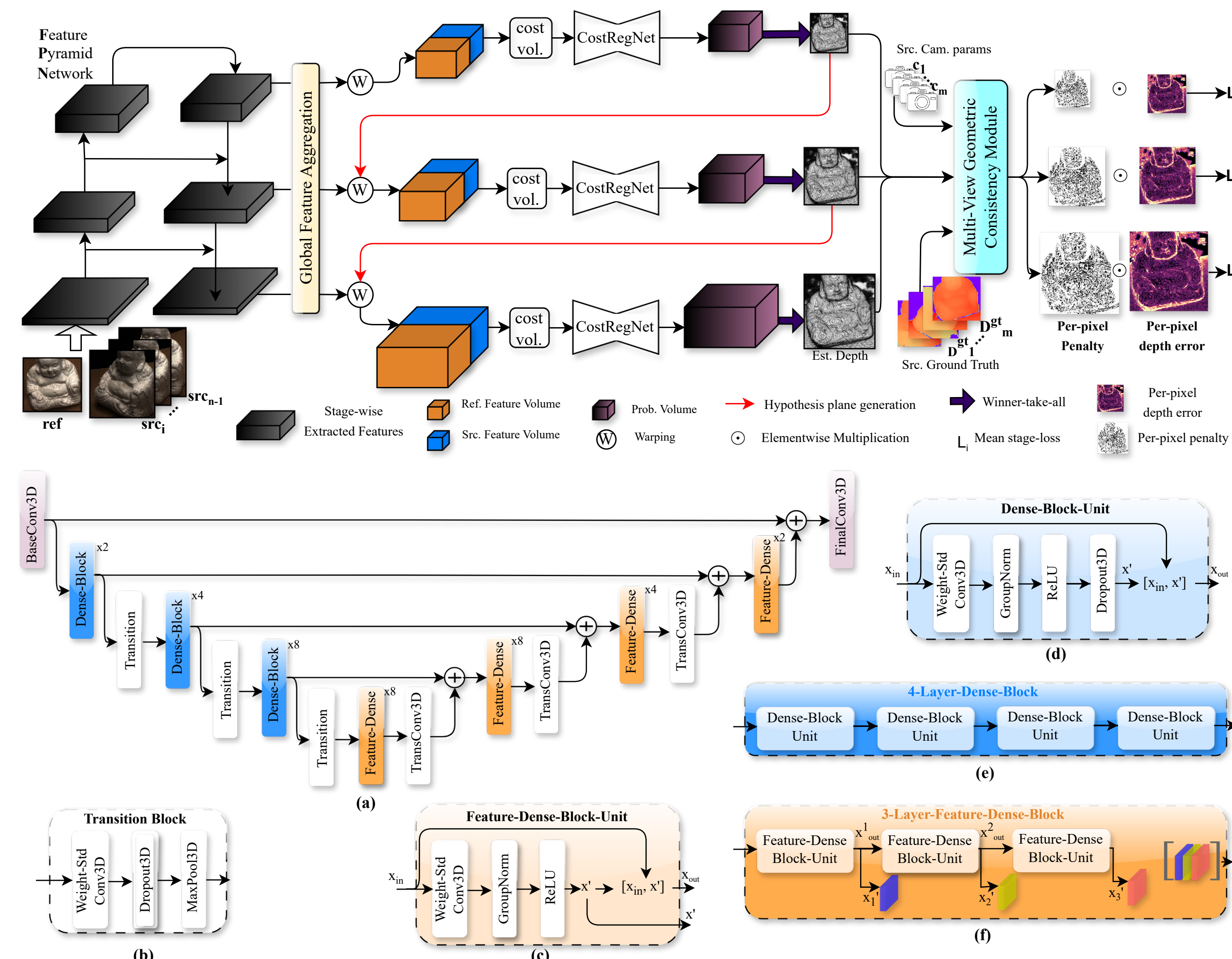
- forward-backward-reprojection to get PDE and RDD
 - $PDE \leftarrow \|P_0 - P_0''\|_2$
 - $RDD \leftarrow \frac{1}{D_0} \|D_{P_0}'' - D_0\|_1$
- Select geometrically inconsistent pixels
 - $PDE_{mask} > D_{pixel}$
 - $RDD_{mask} > D_{depth}$
- Combine inconsistent pixels from both masks
 - Logical-OR (PDE_{mask}, RDD_{mask})
- Current-Mask ← Assign penalty to each pixel
 - Inconsistent pixels → 1
 - All other pixels → 0
- Add Current-Mask to initial Mask-Sum

Geometric penalty (ξ_p) ← average Mask-Sum

Apply reference view binary mask to generate final ξ_p



GC-MVSNet Architecture



Quantitative Results - Tanks & Temples

Our method achieves state-of-the-art results on Tanks & Temples advanced test set and a second place on its intermediate test set.

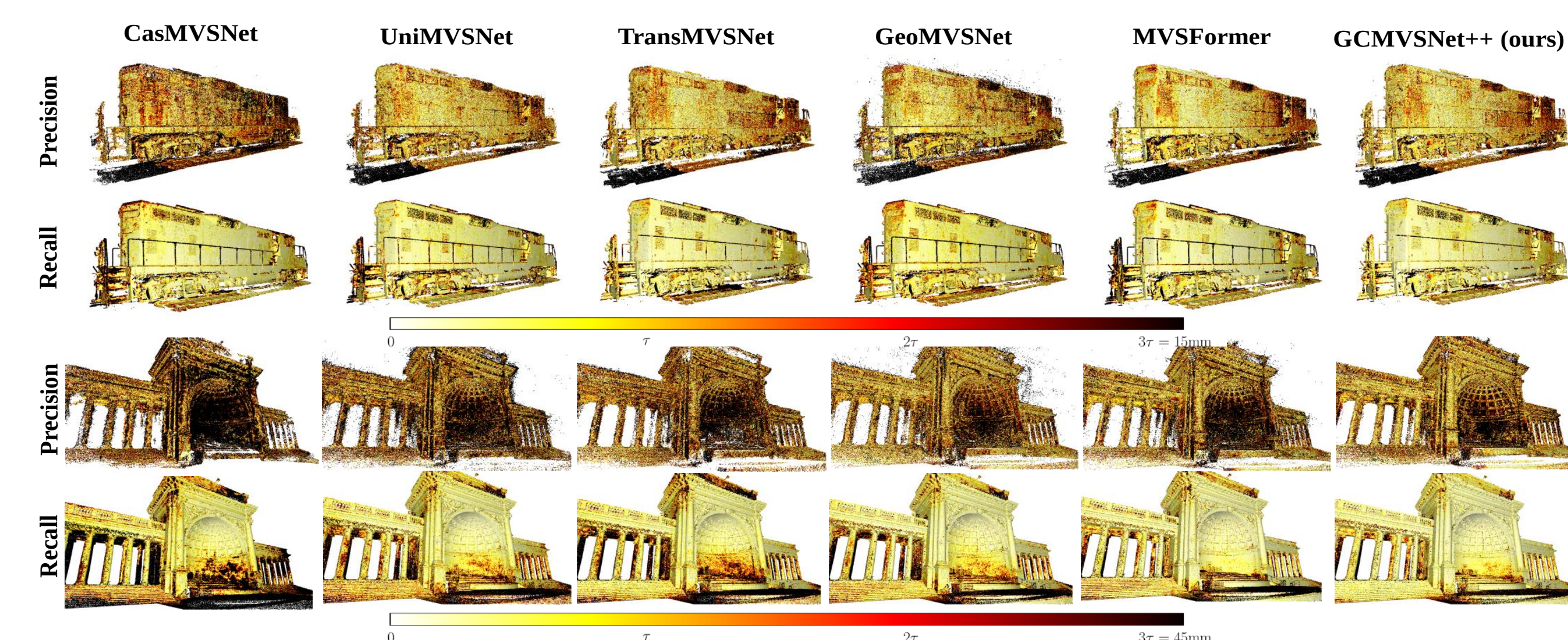
Method	Intermediate set										Advanced set						
	Mean ↑	Fam.	Fra.	Hor.	Lig.	M60	Pan.	Pla.	Tra.		Mean ↑	Aud.	Bal.	Cour.	Mus.	Pal.	Tem.
COLMAP [3]	42.14	50.41	22.25	26.63	56.53	44.83	46.97	48.53	42.04		27.24	16.02	25.23	34.70	41.51	18.05	27.94
R-MVSNet [8]	50.55	73.01	54.56	43.42	43.88	46.80	46.69	50.87	45.25		29.55	19.49	31.45	29.99	42.31	22.94	31.10
CasMVSNet [10]	56.84	76.37	58.45	46.26	55.81	56.11	54.06	58.18	49.51		31.12	19.81	38.46	29.10	43.87	27.36	28.11
AA-RMVSNet [13]	61.51	77.77	59.53	51.53	64.02	64.05	59.47	60.85	54.90		33.53	20.96	40.15	32.05	46.01	29.28	32.71
UniMVSNet [6]	64.36	81.20	66.43	53.11	63.46	66.09	64.84	62.23	57.53		38.96	28.33	44.36	39.74	52.89	33.80	34.63
TransMVSNet [5]	63.52	80.92	65.83	56.94	62.54	63.06	60.00	60.20	58.67		37.00	24.84	44.59	34.77	46.49	34.69	36.62
GBi-Net [60]	61.42	79.77	67.69	51.81	61.25	60.37	55.87	60.67	53.89		37.32	29.77	42.12	36.30	47.69	31.11	36.39
MVSTER [61]	60.92	80.21	63.51	52.30	61.38	61.47	58.16	58.98	51.38		37.53	26.68	42.14	35.65	49.37	32.16	39.19
GeoMVSNet [40]	65.89	81.64	67.53	55.78	68.02	65.49	67.19	63.27	58.22		41.52	30.23	46.53	39.98	53.05	35.98	43.34
MVSFormer [54]	66.37	82.06	69.34	60.49	68.61	65.67	64.08	61.23	59.53		40.87	28.22	46.75	39.30	52.88	35.16	42.95
GC-MVSNet [17]	62.74	80.87	67.13	53.82	61.05	62.60	59.64	58.68	58.48		38.74	25.37	46.50	36.65	49.97	35.81	38.11
GC-MVSNet++ (ours)	66.28	82.72	71.05	58.18	65.40	65.63	64.95	61.47	60.87		42.14	27.74	47.84	40.24	52.32	37.48	47.23

Quantitative Results - DTU and BlendedMVS

Our method achieves state-of-the-art results on two datasets: DTU and BlendedMVS

	Method	DTU Dataset			BlendedMVS Dataset		
		Acc ↓	Comp ↓	Overall ↓	EPE ↓	e_1 ↓	e_3 ↓
Traditional	Furu [2]	0.613	0.941	0.777	—	—	—
	Tola [4]	0.342	1.190	0.766	—	—	—
	Gipuma [1]	0.283	0.873	0.578	—	—	—
	COLMAP [3]	0.400	0.664	0.532	—	—	—
Learning-based	MVSNet [15]	0.396	0.527	0.462	1.49	21.98	8.32
	CasMVSNet [10]	0.325	0.385	0.355	1.43	19.01	9.77
	CVP-MVSNet [11]	0.296	0.406	0.351	1.90	19.73	10.24
	Vis-MVSNet [57]	0.369	0.361	0.365	1.47	15.14	5.13
	AA-RMVSNet [13]	0.376	0.339	0.357	—	—	—
	EPP-MVSNet [65]	0.413	0.296	0.355	1.17	12.66	6.20
	UniMVSNet [6]	0.352	0.278	0.315	—	—	—
	TransMVSNet [5]	0.321	0.289	0.305	0.73	8.32	3.62
	GBi-Net [66]	0.315	0.262	0.289	—	—	—
	MVSTER [67]	0.350	0.276	0.313	—	—	—
	GeoMVSNet [46]	0.331	0.259	0.295	—	—	—
	MVSFormer [60]	0.327	0.251	0.289	—	—	—
	MVSFormer++ [68]	0.309	0.252	0.2805	—	—	—
	GoMVS [69]	0.347	0.227	0.287	—	—	—
	GC-MVSNet [19]	0.330	0.260	0.295	<u>0.48</u>	7.48	2.78
GC-MVSNet++ (ours)		0.319	<u>0.246</u>	<u>0.2825</u>	0.407	5.79	2.41

Error Plot - Train (Tanks & Temples)



GC: A Plug-in Module

The GC module is designed as a plug-in module

- Plug into any depth map-based MVS method
- Retraining the network with GC module provides:
 - Improved quantitative results
 - Requires less training iterations to achieve optimal performance

We demonstrate this on two different methods:

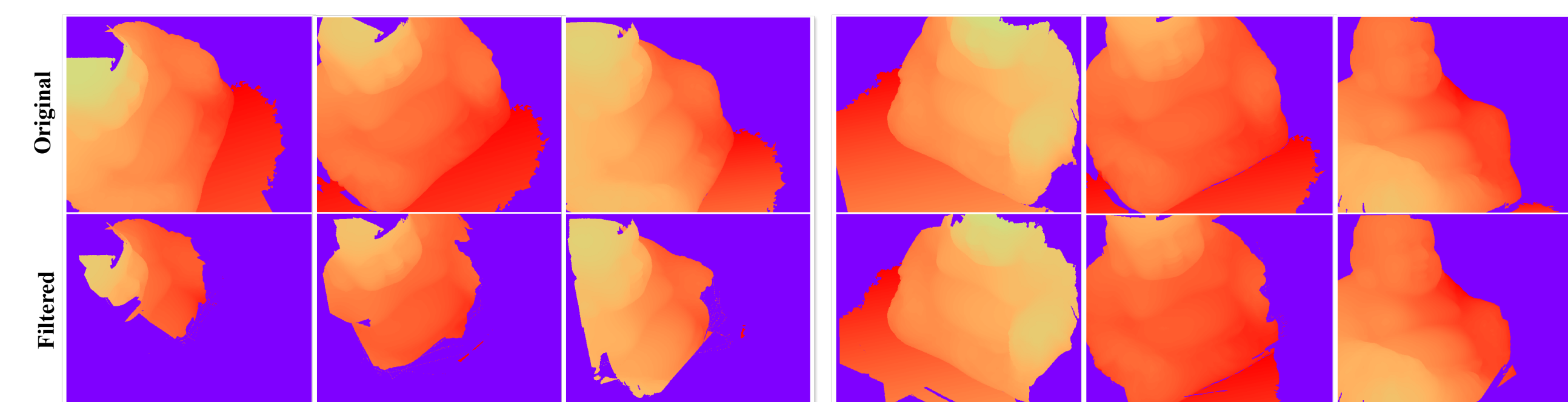
- CasMVSNet and TransMVSNet

Methods	Loss	DepthFiter	Other	GC	Overall↓	Epoch
CasMVSNet [2]	L_1	×	×	×	0.355	16
	L_1	×	✓	×	0.357	16
	L_1	×	×	✓	0.335	11
	L_1	✓	×	×	0.330	11
TransMVSNet [1]	FL	×	×	×	0.305	16
	FL	×	✓	×	0.322	16
	FL	×	×	✓	0.303	8
	FL	✓	×	✓	0.297	8

Depth Filtering

Depth ground truths are inherently noisy. We apply the GC module as a pre-processing step to perform geometry-based depth filtering to improve the quality of depth ground truth.

- GC module removes geometrically inconsistent ground truth depth pixels.
- It requires adjusting two hyperparameters – D_{depth} and D_{pixel} .



References